

Advanced Physics: Mechanics (SCI 514)

Year: 2025 - 2026

Teacher: Luis Luis Fayat (Mr. Luis) - he/him

Room(s): STEM 105 (*Office Hours: STEM 205*)

Contact: lluisfayat@ransomeverglades.org

Course Rationale: Welcome to Advanced Physics: Mechanics. This course is equivalent to a first-semester, algebra-based college physics course. We will explore Newtonian mechanics, including kinematics, forces, energy, momentum, fluids, and waves, through a lens of rigorous conceptual understanding and higher-level mathematics.

Important Note Related to the AP Physics 1 Exam: Please be aware that this course is not designed to prepare students for the AP Physics 1 exam. Although there is some curricular overlap, significant differences exist in the material's breadth, depth, and the specific skills tested on the AP exam. Students are permitted to register for the exam, but should understand that a high degree of independent learning is necessary for success. We encourage students to make a decision that best suits their individual needs and goals.

Scientific Literacy: A key goal of this course is to build scientific literacy. We will explore important real-world applications of physics concepts through hands-on work with data. Laboratory work will focus on developing your ability to design experiments, critically analyze data, and effectively communicate your results.

Grading Distribution & Policies:

Assessments [50%]	All students should aim to always do their best. Each semester may include at least 1 project that covers information from several units, and this will count as a test grade. Tests will have reassessments available to help students work on content mastery.
Labs [20%]	Activities, demos, and labs will be done weekly in this class. Typically, 1-2 labs accompany each unit and are meant to emphasize hands-on skills and a deeper understanding of concepts.
Quizzes [20%]	Check-ins are administered approximately 1-2 times per unit and serve to inform students about their progress in learning and understanding the material.
Assignments [10%]	Assignments will be items and activities that help students focus on their learning. These assignments are a mix of practice problems, review questions, and group work that all serve to give students as many opportunities as they need to feel prepared for assessments as they work towards achieving and maintaining mastery in their learning.

→ Please be prepared to use *myCompass*, *MS Teams* & *OneNote*, *Pivot Interactives*, and *WileyPlus* for this course.

The above sections (which add up to 100%) in reality contribute to **90%** of a student's semester grade. The last **10%** will be determined by the end-of-semester (**EoS**) assessment. The two semesters, along with their respective EoS, weigh equally to determine the final course grade.

Required Textbook: *Physics, Advanced Edition* w/WileyPLUS (11th Edition). Cutnell & Johnson (*e-book only*)

Key Course Tools:

- **OneNote** | Blank notes, labs, handouts, etc., to be distributed often here.
- **Microsoft Teams** | Used to submit assignments, post solutions to practice questions for assessments, etc.
- **Pivot Interactives** | Online and in-class video lab activities.
- **WileyPlus** | Online platform where assignments, reviews, and extra exercises will be assigned.

Materials Required: Please be sure to have these materials available at the beginning of every class:

- 1) Pencils/Pens/Stylus
- 2) Physical notebook to capture notes and work
- 3) Calculator (scientific or graphing)
- 4) Fully charged laptop

Principles of Conduct:

- **Be Honest & Forthright.** Honesty means we are truthful in our words and actions. This creates a foundation of trust where there are "no secrets, no surprises". Commit to academic integrity, take ownership of your mistakes, and communicate openly with your peers and instructor as you strive to improve every day.
- **Be Present & Engaged.** Be conscious of your level of focus and actively participate in the learning environment. This involves minimizing distractions (*specifically your cell phone and laptop*), listening attentively to others, and contributing thoughtfully to discussions and activities. Your engagement is key to both your success and the success of the class.
- **Be Empathetic & Understanding.** Empathy is the ability to understand and share the feelings of others. Practice this by giving others the space to practice, and even struggle, seeking to understand different perspectives when you disagree, and being patient with your classmates, as everyone learns at a different pace.
- **Be Respectful & Considerate.** Showing respect means being considerate of the needs of others in our shared environment. Pay attention and aim to achieve the goals of the class. Value the diverse backgrounds and opinions in the classroom, be thoughtful with your language, and handle shared materials and lab equipment with care.
- **Be Grateful & Positive.** Gratitude encourages us to be positive and take perspective, even when facing challenges. Acknowledge the effort of your peers and instructor, appreciate the opportunity to learn, and approach new topics with an open and optimistic mindset

Missed Assessments and Assignments:

To ensure fairness and maintain academic momentum, the following policy applies to any missed assessments:

- **Planned Absences:** If a student knows in advance that they will miss a test, the student must notify the instructor beforehand to arrange an appropriate makeup date and time.
- **Unexpected Absences:** In the event of an unexpected absence, such as an illness, the student should notify the instructor as soon as possible. For any excused absence, the instructor and student will work together to schedule a makeup test at a mutually agreeable time.
- **Lack of Communication:** A missed test without prior notification will be made up at a time and place arranged by the instructor. Failure to keep a scheduled makeup appointment without notifying the instructor may result in a penalty.

Assignments submitted after the due date without an agreed-upon extension will be subject to a 10% late penalty. Additional deductions may apply if the assignment remains incomplete after the extended deadline. The minimum grade a student will receive for a late but completed assignment that meets basic expectations is 50%.

Writing and Technology Expectations: To practice effective scientific communication, students will complete various technical assignments such as research papers and lab reports. This requires proficiency in both the Microsoft and Google Suites, as well as familiarity with the basic functions of a scientific calculator.

Important Note on Cell Phone Use in Class: To maintain a focused learning environment, all student phones will be collected and securely stored at the beginning of class and returned at the end of the period.

Test Reassessments: If a student receives a grade below a "C" (73) on an assessment, they have an optional opportunity for reassessment. The goal of this policy is to help students close any gaps in their understanding and ensure that a single poor grade does not make it difficult for them to recover academically.

To be eligible, the student must:

1. Schedule an appointment to meet with the instructor **within one week** of the assessment being returned.
2. Bring evidence of extra preparation to the meeting, such as completed quiz corrections or extra practice problems, to show that the student has reviewed their mistakes from the exam.

After the student and instructor discuss the material and mutually agree that the student's understanding has improved, the student's grade on the assessment will be raised to a 73.

Cheating and Academic Dishonesty: Any student caught cheating will have the assignment confiscated, and the incident will be reported in accordance with the student handbook. If you are struggling, please speak with the instructor for support instead of resorting to cheating.

Safety and Orderliness: A signed “Flinn Scientific Student Safety Contract” is required for participation in all classroom demonstrations, activities, and labs. Appropriate behavior is mandatory for a safe learning environment.

→ This course contract includes the guidelines and requirements for this course. Please remember to use it as a reference throughout the year, and note that all school rules and policies must be adhered to in this class.

AI Policy: To prepare students for the ubiquity of AI in schools, colleges, and other aspects of life, some courses and curricula at RE incorporate and encourage the innovative use of AI. To facilitate clear faculty-to-student communication around the responsible use of AI, teachers will code all assignments using the following three-tier system:

Red: Human-Only Zone: AI use is not permitted for any aspect of the assignment.

Yellow: Guided Use Only: AI may be used in an assistant role, for certain components of the assignment as clearly defined by the instructor.

Green: Open Use with Accountability: AI may be used in a collaborator or thought-partner role, with full transparency and acknowledgment.

Advanced Physics: Mechanics - Tentative Course Topics:

Conversions & Measurements	Momentum & Collisions
Vectors and Scalars	Simple Harmonic Motion
Forces and Newton’s Laws	Circular Motion & Gravitation
Kinematics (1-D & 2-D)	Fluids
Work & Energy	Waves & Sound

**Advanced Physics: Mechanics Course Contract
2025 - 2026
Mr. Luis**

We, the below signed, have read, understand, and will adhere to the guidelines and requirements addressed in the Advanced Physics: Mechanics Course Packet.

Student’s Name: _____

Student’s Signature: _____

Parent’s Name: _____

Parent’s Signature: _____

*Parent’s Name: _____

Parent’s Signature: _____

Date: _____

**Please Note: Only 1 parent signature is required, but I would encourage all parents to read through this packet if possible / applicable.*